each operator must cover the entire area or have agreements in place to permit roaming.

Second to Third Generations Bridge (2.5G) 2000 – GPRS launch

GPRS was introduced here, this is the excess period of mobile networking development, between 2G and 3G. GPRS stands for(**General Packet Radio Service**).GPRS is a data service which enables mobile devices to send and receive messages, picture messages and e-mails. It allows most popular operating speeds of up to 115kbit/s, latterly maximum of 384kbit/s by usingEDGE. EDGE stands for **EDGE (Enhanced Data rates for Global Evolution)**. GSM data transmission rates typically reached 9.6kbit/s.

Third Generation(3G)2003 – First UK 3G launch

This generation of mobile system merges different mobile technology standards, and uses higher frequency bands for transmission and Code Division Multiple Access to deliver data rates of up to 2Mbit/s supporting multimedia services (MMS: voice, video and data). European standard is UMTS **(Universal Mobile Telecommunication Systems).** Mobile phones systems continue to use digital transmission with SIM authentication for billing systems and for data incorruption. Data transmission used a WCDMA. WCDMA stands for (Wideband Code Division Multiple Access). A technique to obtain data rates between 384kbit/s and 2048kbit/s. Few 3G suppliers use ATM (Asynchronous Transfer Mode) for '**over the air**' network with in MPLS (Multiprotocol Label Switching) or IP for their backbone network.

Mobility still supported at layer 2, and hence like 2G it still prohibits seamless roaming beyond heterogeneous access networks and routing domains. The transmission were band frequencies between 1900 and 2200 MHz. All UMTS license holders at the UK holds a 20 year license with the condition that 80% population coverage is achieved by 31 December 2007. The present third generation licensed operators in the UK can be seen below as at August 2004.

Fourth Generation(4G) 2007

4G is at the research stage. 4G was based on an adhoc networking model where there was no need for a fixed infrastructure operation. Adhoc networking requires global mobility features (e.g. Mobile IP) and connectivity to a global IPv6 network to support an IP address for each mobile device. Logically roaming in assorted IP networks (for example: 802.11 WLAN, GPRS and UMTS) were possible with higher data rates, from 2Mbit/s to 10–100Mbit/s, offering reduced delays and new services. Mobile devices will not expect on a fixed infrastructure, they



Li-Fi is a wireless technology which uses light-emitting diodes (LEDs) for data transmission whereas Wi-Fi uses radio frequencies for data transmission. Li-Fi is the short form of Light Fidelity.

The term Li-Fi was first used by Harald Haas, Professor in Edinburgh University. The computer scientists achieved speeds of 224 gbps in the lab and research is going on. The biggest revolution in the Internet world is going to happen

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